

Buyu Deng 邓卜瑜

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🌐 <https://buyudeng.github.io/>

🔗 BuyuDeng



Education

2023.09 – Expected **B.S. Candidate**, *School of the Gifted Young*, University of Science and Technology of China (USTC)
2027.06 **Major:** Geophysics

GPA: 3.62/4.30 (**Rank:** 3/20)

Relevant Coursework: Linear Algebra: **90**, Probability & Statistics: **92**, Theoretical Mechanics: **95**, Python & Deep Learning: **A**, Programming in Geophysics: **94**, Introduction to Seismic Exploration: **98**

Research Interests: Seismic interpretation, geomodeling, forward modeling; deep learning, agentic AI

Research Experience

2026.07 – Present **Undergraduate Research Intern**, [Stanford Center for Earth Resources Forecasting \(SCERF\)](#), Stanford University
Supervisor: **Prof. Tapan Mukerji**

- Conduct research on geomodeling and forward modeling for subsurface energy systems.

2024.08 – Present **Undergraduate Researcher**, [Computational Interpretation Group](#)
Supervisor: **Prof. Xinming Wu**

3D "All-in-One" Seismic Denoising

- Built a synthetic 3D seismic dataset with random noise, acquisition footprints, and migration artifacts; trained an all-in-one denoising network based on Degradation Representation Learning.
Code: github.com/BuyuDeng/NDR_Seismic

Geo-VAE for 3D Geophysical Data Compression & Latent Processing

- Adapted Wan-VAE into a unified Geo-VAE for seismic volumes, RGT cubes, property models, structures, and geobodies, achieving $8 \times 8 \times 8$ volumetric compression with high-fidelity reconstruction.
- Enabled memory-efficient layer-by-layer inference on large 3D volumes using causal 3D convolutions and feature caching, avoiding chunk-boundary artifacts.
- Built latent-space downstream workflows for missing-trace interpolation with a dual-mask latent diffusion model and seismic denoising with an HRNet-style latent mapper.
Full manuscript under revision at *Geophysics*; gave an oral presentation at IMAGE 2026.
Code: github.com/BuyuDeng/Geo-VAE

PHY-to-BGC Ocean Biogeochemical Emulation

- Built a 2.5D full-image U-Net emulator mapping global physical ocean fields to coupled biogeochemical variables, with evaluation under anomalous ocean-climate conditions; **manuscript in preparation**.

GNSS-NISAR Ionospheric Super-Resolution Reconstruction

- Developed an automated pipeline aligning GNSS TEC-derived ionospheric phase with NISAR dual-frequency InSAR for high-resolution super-resolution research.

Honors and Awards

- 2025 **Zhao Jiuzhang Elite Program Scholarship**, School of Earth and Space Sciences
2024 **Rose Fund Endeavor Scholarship**, School of the Gifted Young
2024 **Zhao Jiuzhang Elite Program Scholarship**, School of Earth and Space Sciences
2023 **Freshman Scholarship**, University of Science and Technology of China (USTC)

Experience and Leadership

2025.08 **IPGP Geophysical Field Camp**: conducted geophysical surveys and joined academic seminars with faculty from the Institut de Physique du Globe de Paris (IPGP).

2025.06 – 2026.01 **Head (Minister)**, Dept. of Cultural & Sports in College Student Union